

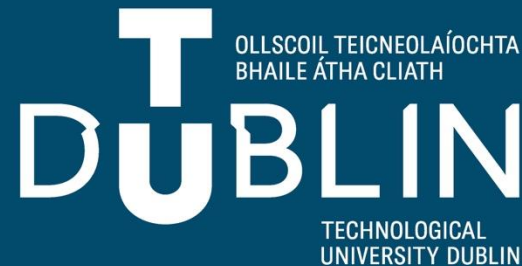
Féidearthachtaí as Cuimse
Infinite Possibilities

Machine Learning

Lecture 12: Unsupervised Learning

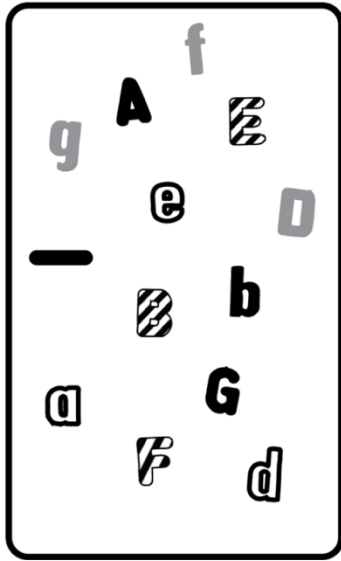
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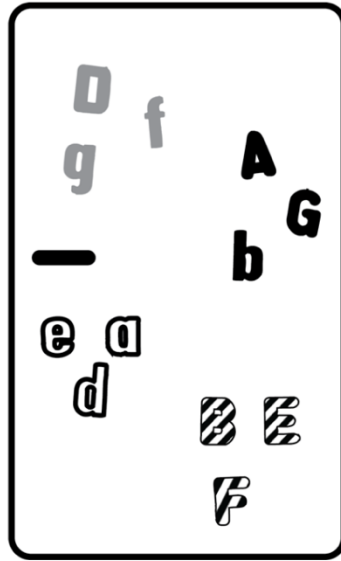


Slides adapted from Sarah Jane Delany and book slides from: Fundamentals of Machine Learning for Predictive Data Analytics.
Kelleher, Mac Namee and D'Arcy

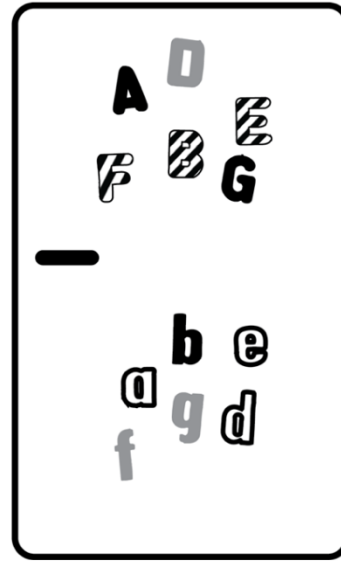
Big Idea



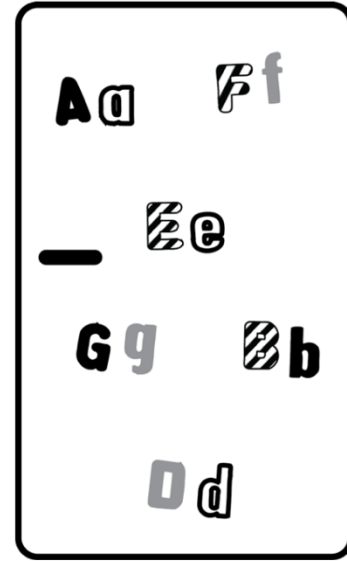
(a) The fridge



(b) Abigail



(c) Andrew

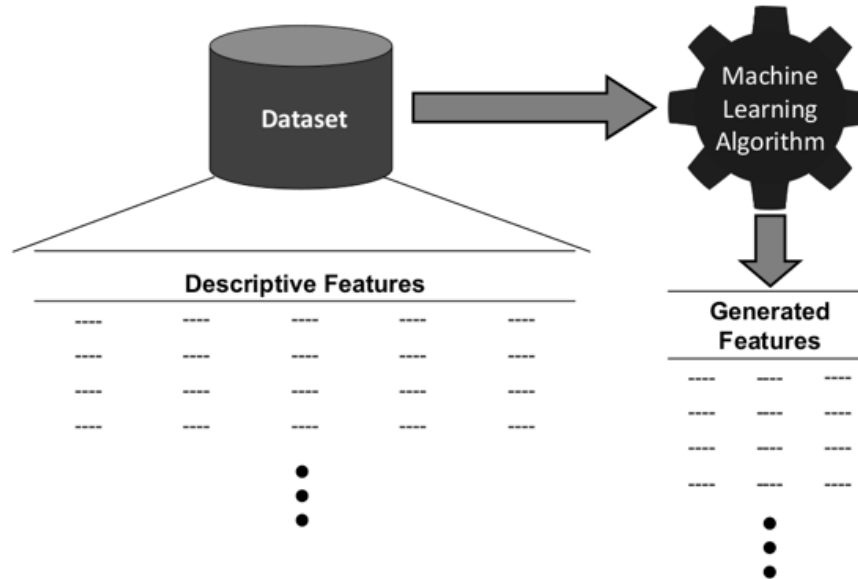


(d) Amalia

- Arrangements of letters on a fridge by three different kids - which is better?
- Difference with supervised learning - no target feature

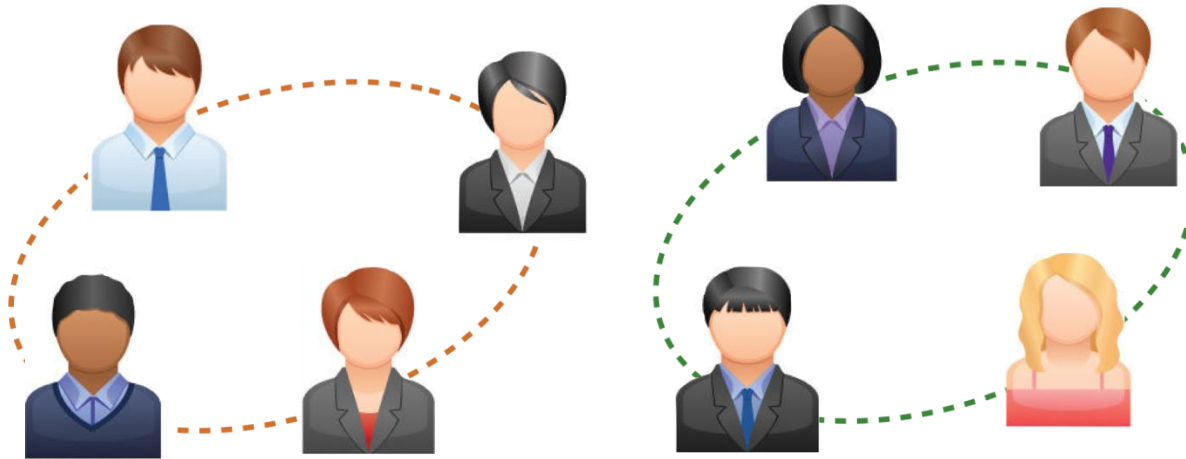
Unsupervised Learning

- Objective: find structure within a set of instances defined by descriptive features alone.
- Structure is typically captured in new generated features that can be appended to the original dataset to **augment** or **enrich** it.



Applications

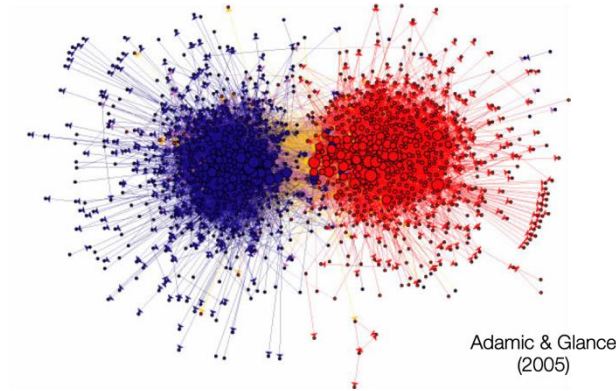
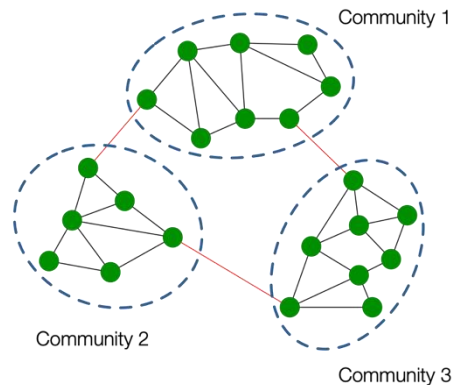
Perform **market segmentation** by grouping customers into separate clusters, so that customers in the same cluster have similar characteristics.



Clustering algorithm output can support targeted marketing and operations for specific sub-groups of customers.

Applications

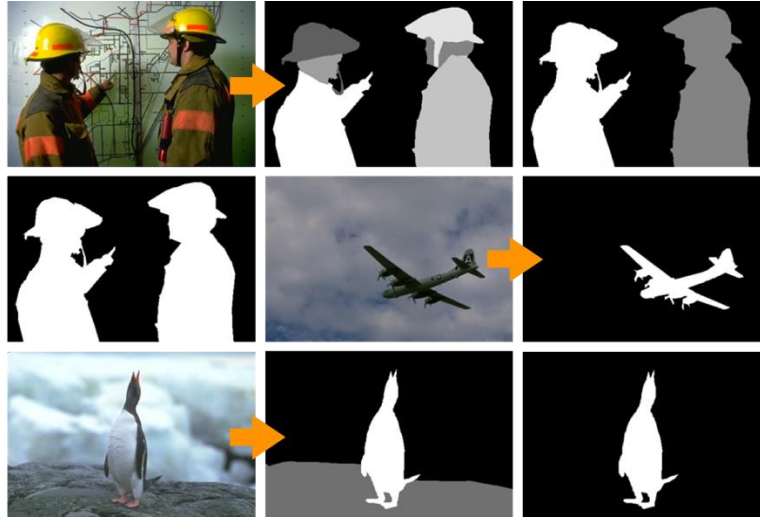
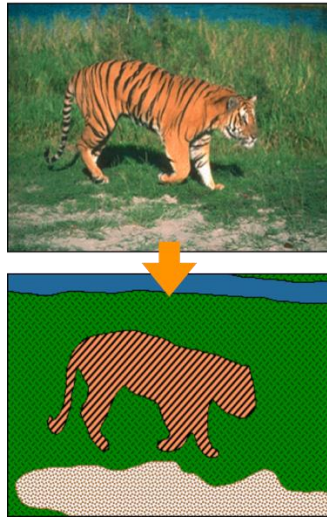
Given a social network, apply unsupervised algorithms to identify **communities** of users who are well-connected to one another, and who are separated from other communities.



Users in the network do not need to be manually labeled or annotated in order to apply the community finding algorithm.

Application

Image segmentation: Unsupervised task in computer vision that attempts to automatically split an image into regions with similar colour or texture, or both. Aim is to partition the image into its constituent “objects”.



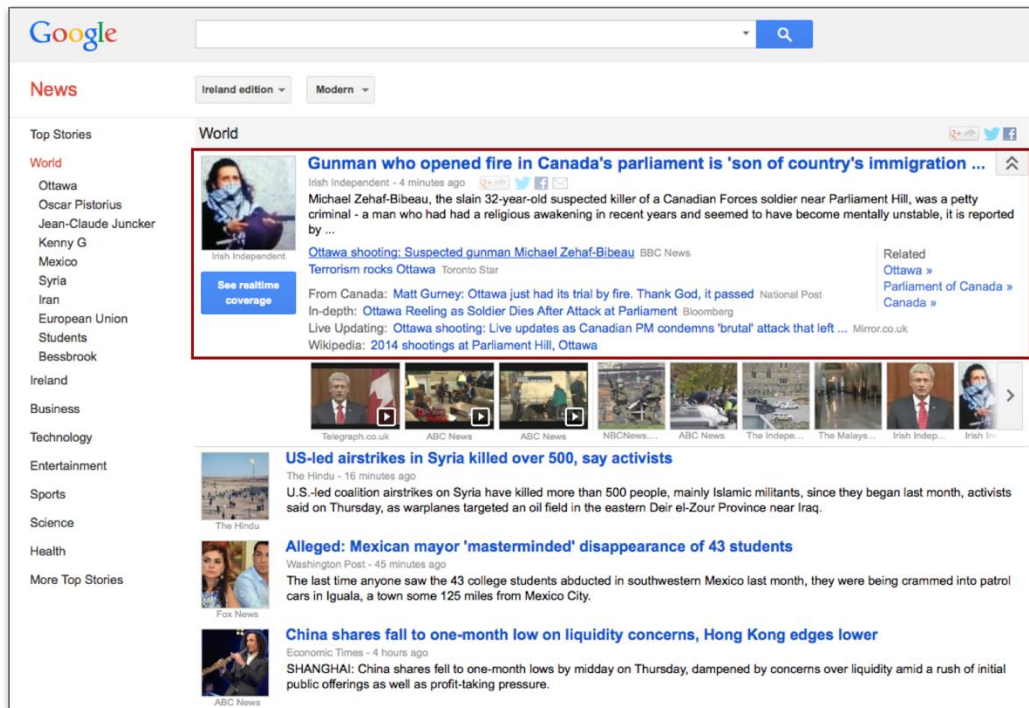
Applications

Topic modeling: Unsupervised task of discovering the underlying thematic structure in a text corpus - i.e. the key “topics” in the data.



Application

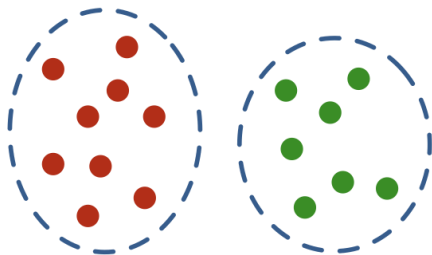
Document clustering: Automatically group related documents together based on similar content (e.g. related articles on Google News).



Clustering

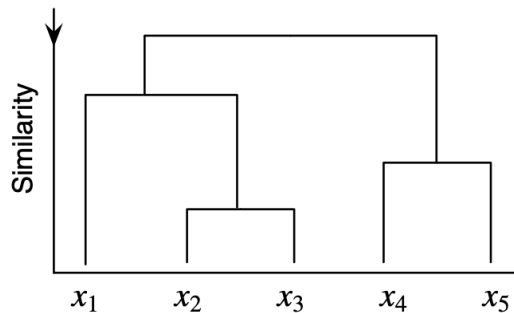
- **Clustering** partitions a dataset into groups or clusters that are similar to each other
- Algorithms use definitions of similarity/dissimilarity and objective functions for determining a “good” cluster

Partitional Algorithms



Build a “flat” clustering of the data all at once

Hierarchical Algorithms



Gradually build a nested tree structure of clusters

Partitional Clustering

- Attempt to directly decompose a data set into a “flat” grouping consisting of a number of disjoint (non-overlapping) clusters.
- Usually pre-specify number of clusters k , although some methods adaptively add/remove clusters.
- Start with an initial set of k clusters, often chosen at random.
- Use a heuristic to find the best local solution for an objective function, identified by iteratively improving the initial solution.
- **k -means** clustering is the most well-known approach to partitional clustering
 - + simple, computationally efficient, quite effective
 - non-deterministic

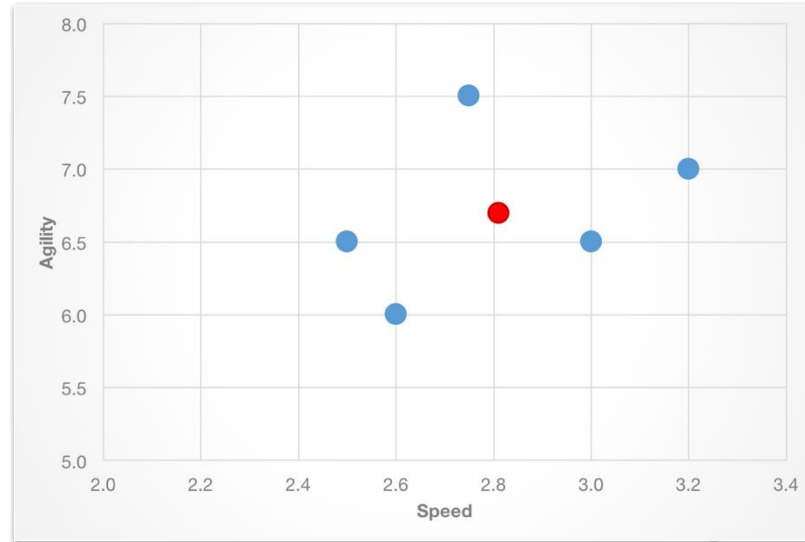
k-means Clustering

Centroid: The mean vector of all items assigned to a given cluster (i.e. the mean of their feature vectors).

<i>Athlete</i>	<i>Speed</i>	<i>Agility</i>
1	2.6	6.0
2	3.0	6.5
3	2.5	6.5
4	3.2	7.0
5	2.8	7.5
Centroid	2.82	6.7

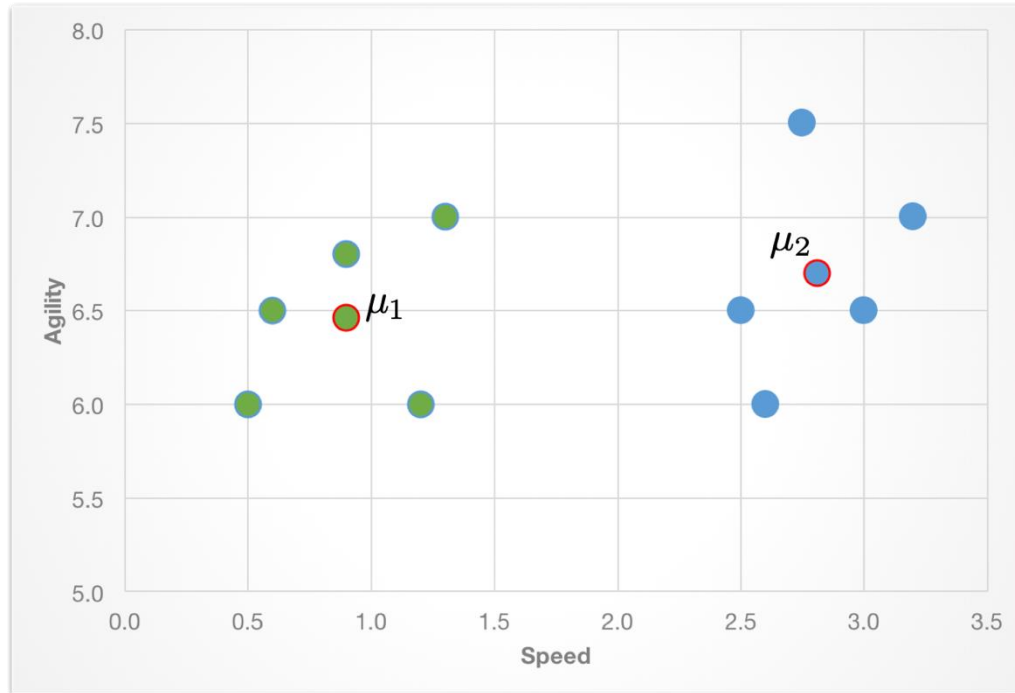
$$(2.6 + 3.0 + 2.5 + 3.2 + 2.8)/5 \\ = 2.82$$

$$(6.0 + 6.5 + 6.5 + 7.0 + 7.5)/5 \\ = 6.7$$



k -means Clustering

- Each of the k clusters in a clustering can be represented by its own centroid μ_i
- Example of two clusters, with centroids shown:



k -means Clustering

For dataset D with instances, $d_1 \dots d_n$ goal is to minimise the distance between instances and their nearest centroid, i.e. minimising the sum-of-squared error (SSE):

$$SSE = \sum_{c=1}^k \sum_{d_i \in C_c} \text{Dist}(d_i, \mu_c)^2 \text{ where } \mu_c = \frac{\sum_{d_i \in C_c} d_i}{|C_c|}$$

Distance is normally measured using Euclidean distance

$$\text{Dist}(d_i, \mu_c) = \sqrt{\sum_{j=1}^m (d_{ij} - \mu_{cj})^2}$$

sum of squared difference
over all m feature values

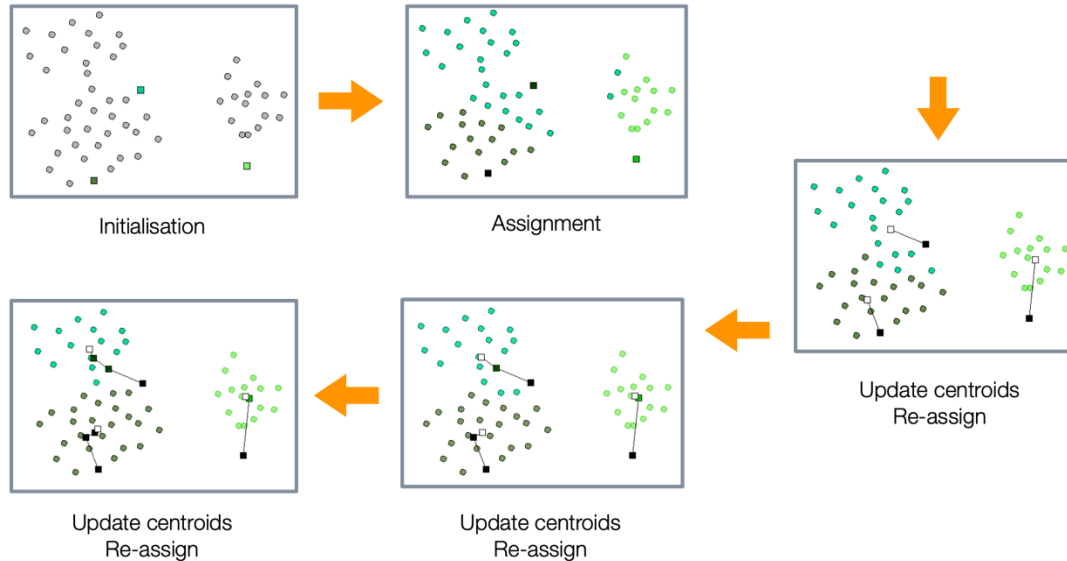
k -Means tries to reduce SSE via a two step iterative process:

- 1) Reassign items to their nearest cluster centroid
- 2) Update the centroids based on the new assignments

Repeatedly apply these two steps until the algorithm converges (i.e. no cluster reassignments happen)

k -means Clustering

Simple example of several iterations of k -Means for $k=3$...



k -means Algorithm

Inputs:

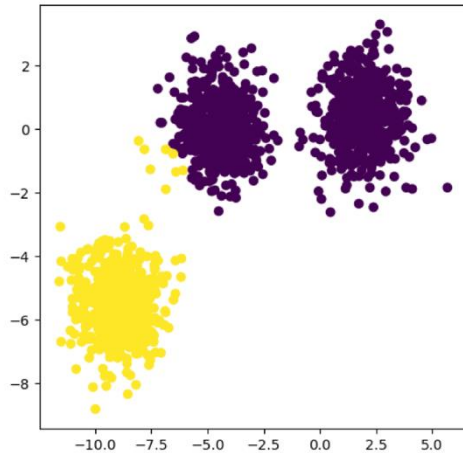
- Data: Set of unlabelled items
- k : User-specified target number of clusters
- Maximum number of iterations to run

Algorithm Steps:

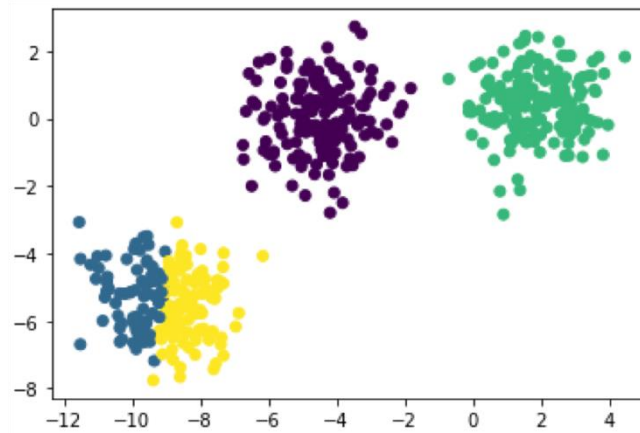
1. *Initialisation*: Select k initial cluster centroids (e.g. at random)
2. *Assignment step*: Assign every item to its nearest cluster centroid (e.g. using Euclidean distance).
3. *Update step*: Recompute the centroids of the clusters based on the new cluster assignments, where a centroid is the mean point of its cluster.
4. Go back to Step 2, until no reassignments occur (or until a maximum number of iterations is reached).

k -means: How many Clusters?

- Key input parameter k - how many clusters?
- k too low $\rightarrow$ “smearing” of clusters that should not be merged.
- k too high $\rightarrow$ “over-clustering” of the data into many small, similar clusters.



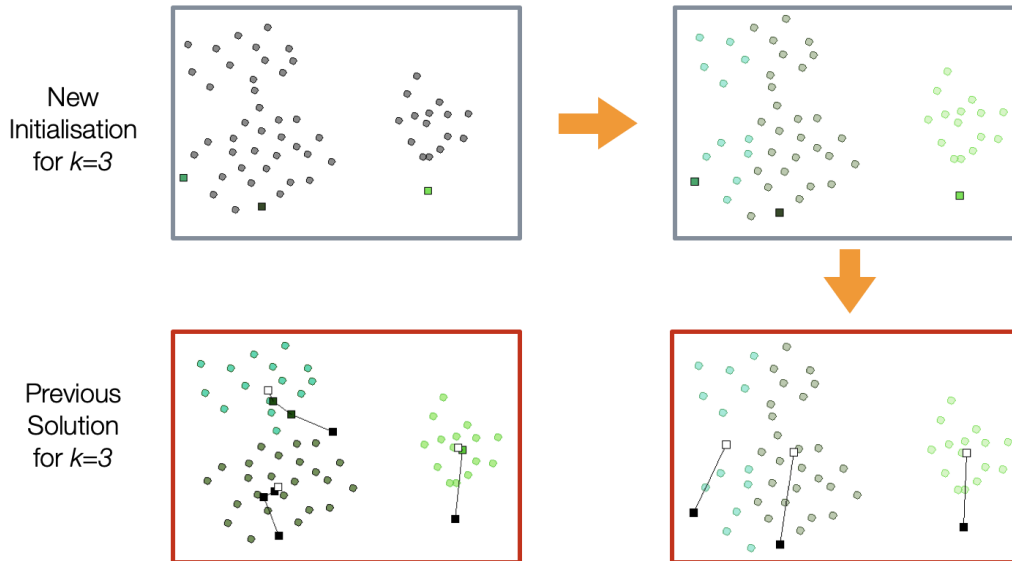
Too few clusters?



Too many clusters?

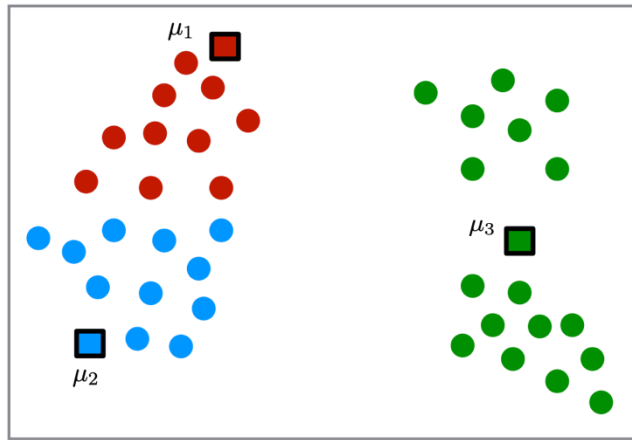
Cluster Initialisation

- Results produced by k -Means are often highly dependent on the initialisation.
- Different starting positions can lead to different local minima - i.e. different clusterings of the same data → non-deterministic

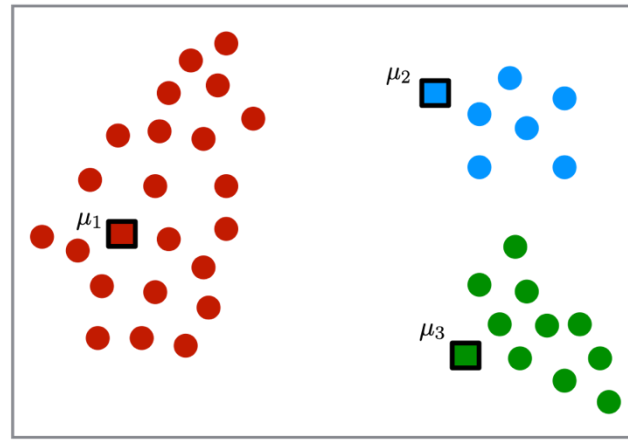


Cluster Initialisation

A poor choice of initial centroids will often lead to a poor clustering that is not useful.
A better initialisation will lead to different clusters.



Initialisation 1



Initialisation 2

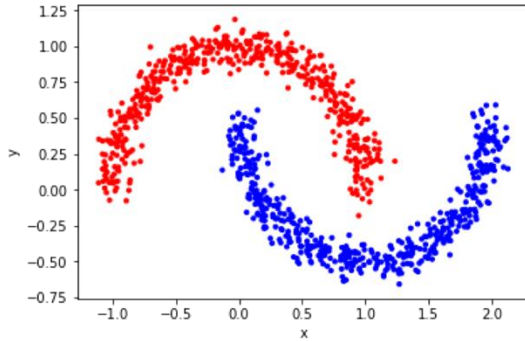
- Common strategy: Run the algorithm multiple times, select the solution(s) that scores well according to some validation measure.

Limitations of k -means

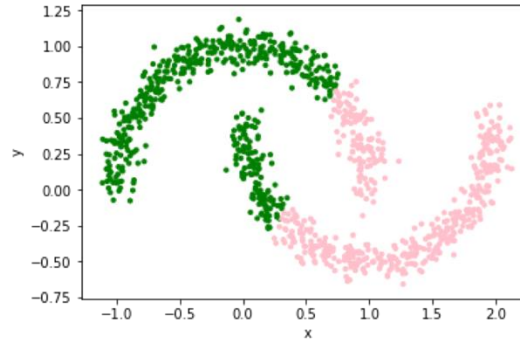
- **Advantages:**
 - Fast, easy to implement.
 - “Good enough” in a wide variety of tasks and domains.
- **Disadvantages:**
 - Must pre-specify number of clusters k .
 - Highly sensitive to choice of initial clusters.
 - Assumes that each cluster is spherical in shape and data examples are largely concentrated near its centroid.
 - Traditional objective can give undue influence to outliers.
 - Iterative process can lead to empty clusters, particularly for higher values of k .

Limitations of k -means

Example: k -Means assumes that clusters are spherical in shape and data examples are largely concentrated near its centroid.



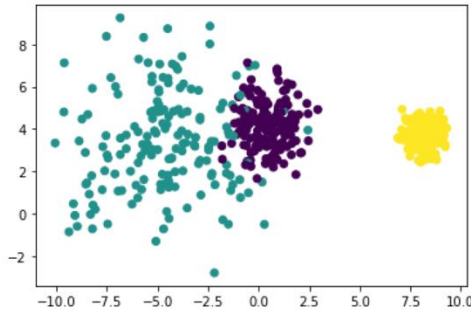
Original “correct” groups in the data



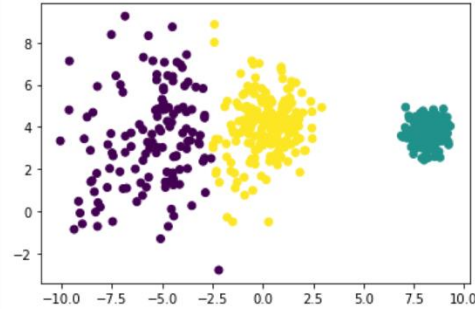
Clusters identified by k -means for $k=2$

Limitations of k -means

Example: k -Means assumes that clusters are spherical in shape and data examples are largely concentrated near its centroid.



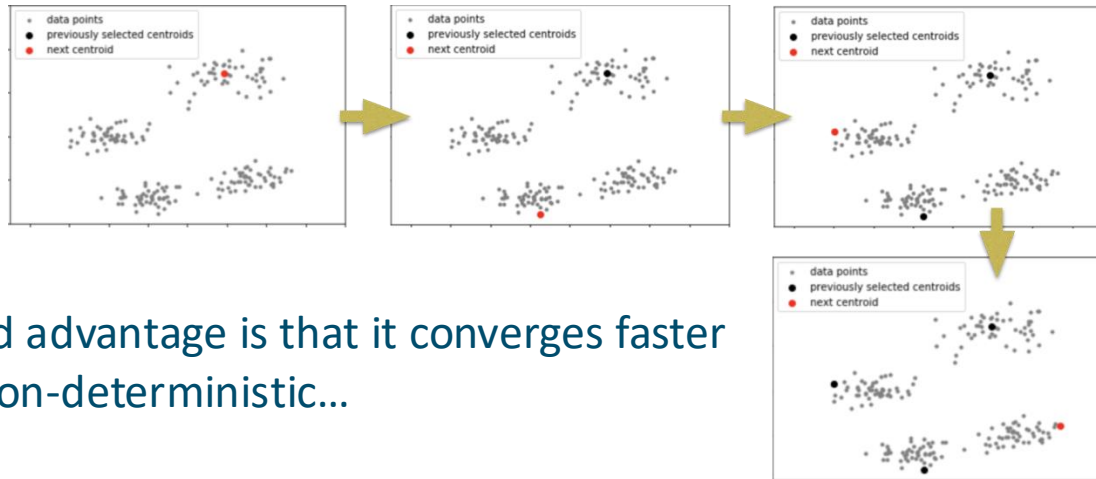
Original “correct” groups in the data that has clusters of varied density



Clusters identified by k -means for $k=3$

k -means++

- k -means++ is a variation on k -means that includes an initialisation strategy that selects initial centroids less likely to lead to sub-optimal clusters
- Instances far away from the already selected centroids are more likely to be selected than those close to the already selected centroids



- Added advantage is that it converges faster
- Still non-deterministic...

k-means++

Algorithm:

1. Randomly select first centroid from the dataset $\{d_1 \dots d_n\}$
2. For each instance d_i in the dataset calculate its distance from its nearest centroid
3. Assign a weight w_i to each instance d_i which is proportional to the square of this distance

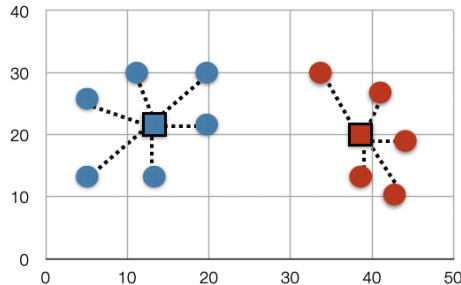
$$w_i = \frac{Dist(d_i)^2}{\sum_{p=1}^n Dist(d_p)^2}$$

4. Select the next centroid randomly from the dataset with each instance weighted by w_i
5. Repeat from 2 until k centroids are selected

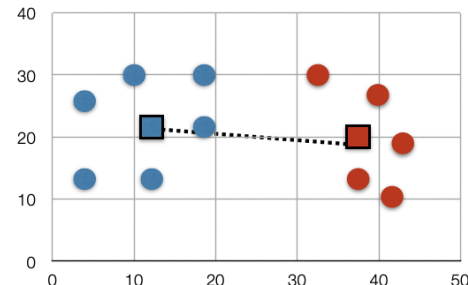
Proceed with *k*-means using selected centroids as the initial centroids

Evaluating Clusters

- **Cluster validation**: Measures for automatically producing a quantitative evaluation of the quality of a clustering.
- Common motivation - “good” clusters have the property that cluster members are close to each other and far from members of other clusters
- Good clustering **minimises intra-cluster distances** (aka **inertia**) and maximises **inter-cluster distances**



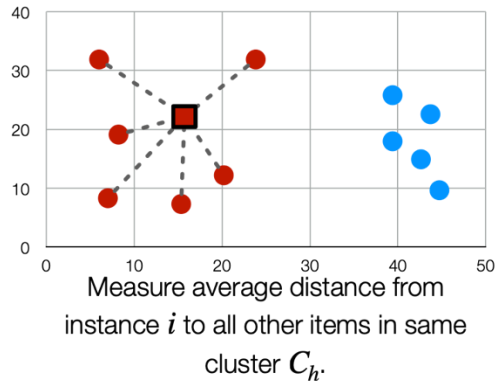
Intra-cluster distance



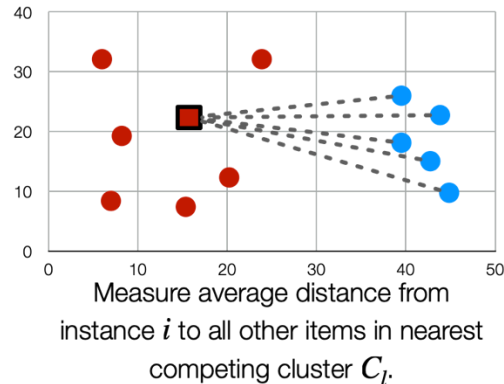
Inter-cluster distance

Silhouette Measure

- Validation measure which quantifies degree to which each item belongs in its assigned cluster, relative to the other clusters.



$$a_i = \frac{1}{|C_h| - 1} \sum_{j \in C_h, j \neq i} d(i, j)$$



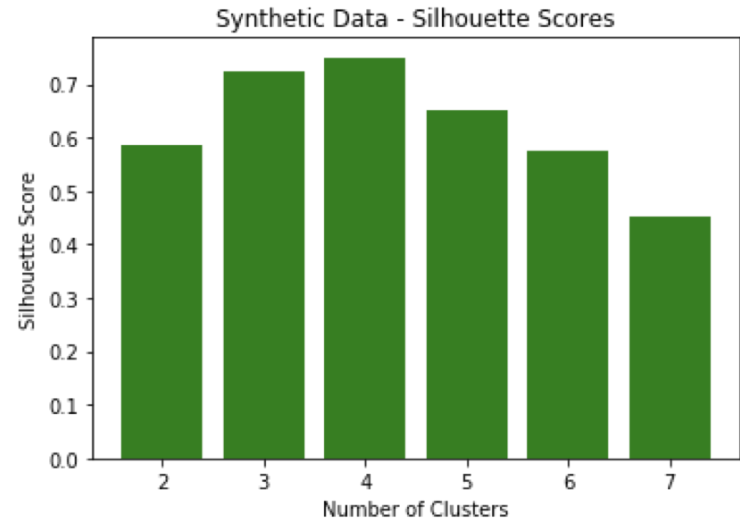
$$b_i = \frac{1}{|C_l|} \sum_{j \in C_l} d(i, j)$$

- Silhouette width** for an item x_i is given by s_i . Values are in the range $[-1, 1]$, a larger value is better.

$$s_i = \frac{b_i - a_i}{\max \{a_i, b_i\}}$$

k-means: How many Clusters?

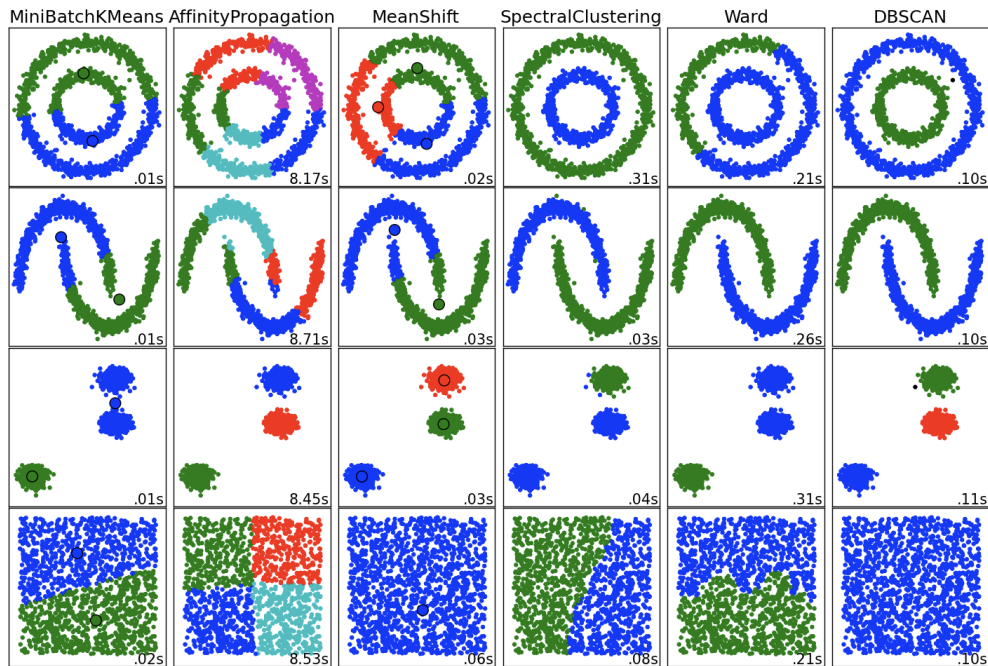
- Cluster validation is often applied for parameter selection - e.g. select an appropriate value k for the k -Means algorithm.
- **Typical Strategy:**
 - Apply k -Means for each value from $k_{\min}$ to $k_{\max}$.
 - Calculate score for each clustering using a cluster validation measure.
 - Examine plot of scores to identify a peak for the best value for k .



Clustering

There are usually many different ways to cluster the same data. Clustering algorithms differ significantly in their definition of what constitutes a cluster and the approach used to find them.

<http://scikit-learn.org/stable/modules/clustering.html>



Questions?